

Gaussian Rigidity from Symmetrized Moment Generating Functions: Chi-Square Quadratic Forms and Sharp Reflection Stability

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Abstract

We study Gaussian rigidity through the symmetrized moment generating function $M_{\mathbf{X}}(\mathbf{s})M_{\mathbf{X}}(-\mathbf{s})$. For independent centered coordinates $\mathbf{X} = (\mathbf{X}_i)$ and a positive-semidefinite matrix $\mathbf{A}$, we prove that if $\mathbf{X}^\top \mathbf{A} \mathbf{X}$ has a chi-square distribution and every active coordinate satisfies the local dominance $M_i(\mathbf{s})M_i(-\mathbf{s}) \geq \exp(\sigma_i^2 \mathbf{s}^2)$, then all active coordinates are Gaussian. We then turn to the quantitative factor-recovery problem. If $\mathbf{X}$ is centered with variance one and

$$\log M_{\mathbf{X}}(\mathbf{s}) + \log M_{\mathbf{X}}(-\mathbf{s}) - \mathbf{s}^2$$

is uniformly bounded in absolute value by Δ on a fixed neighborhood of the origin, then

$$d_{\mathbf{K}}(\mathbf{X}, \mathcal{N}(\mathbf{0}, \mathbf{I})) \leq C \sqrt{\frac{\log \log(1/\Delta)}{\log(1/\Delta)}}.$$

No separate exponential-moment hypothesis is required: the local reflected-MGF bound itself yields the exponential envelope used in the proof. The order is optimal even within symmetric densities uniformly comparable with the Gaussian density, as shown by an even Laguerre perturbation. The proof converts local reflected-MGF information into high-order moments, produces a growing zero-free disk for a truncated characteristic function, and then applies logarithmic coefficient bounds and Esseen smoothing. As applications, we obtain exact and quantitative results for the sample variance, including a stability corollary stated directly in Kolmogorov distance from the chi-square law; a lattice example shows that no analogous total-variation stability can hold.

Keywords: Gaussian characterization; symmetrized moment generating function; chi-square quadratic forms; sample variance; Cramér stability; zero-free characteristic functions

1 Introduction and main results

The Gaussian decomposition principle used in the factor-recovery argument is the classical theorem of Cramér [1].

The exact chi-square law of the sample variance is one of the classical signatures of Gaussian samples. The unrestricted converse for a fixed sample size $n \geq 3$ remains a longstanding characterization problem. Existing results impose additional structure, including symmetry or infinite divisibility; see [2–7]. We do not claim to resolve that unrestricted problem. The sample variance is a chi-square quadratic form, and spherical averaging of its Laplace transform naturally brings in the symmetrized quantity $M_X(s)M_X(-s)$.

For a centered variance-one random variable X , let $M_X(s) = \mathbb{E}e^{sX}$ whenever finite and define

$$R_X(s) := \log M_X(s) + \log M_X(-s) - s^2, \quad \Delta_\rho(X) := \sup_{|s| \leq \rho} |R_X(s)|. \quad (1)$$

We write $d_K(X, Z)$ for the Kolmogorov distance between the laws of X and Z ,

$$d_K(X, Z) := \sup_x |\mathbb{P}(X \leq x) - \mathbb{P}(Z \leq x)|.$$

If X' is an independent copy of X , then $M_{X-X'}(s) = M_X(s)M_X(-s)$. The difficulty is that reflection removes the Fourier phase. Our stability theorem shows that the factor can nevertheless be recovered, at the optimal logarithmic scale, from local real-axis information alone.

A useful self-improvement is that the reflected defect already supplies the exponential integrability needed later. Indeed, if $\mathbb{E}X = 0$, then Jensen's inequality gives $M_X(s) \geq 1$. Hence, whenever M_X is finite on $[-\rho, \rho]$ and $\Delta_\rho(X) \leq \Delta$,

$$M_X(\rho) + M_X(-\rho) \leq 2e^{\rho^2 + \Delta}, \quad \mathbb{E}e^{\rho|X|} \leq 2e^{\rho^2 + \Delta}. \quad (2)$$

Thus no independent exponential-moment assumption is needed in the reflection theorem.

The proof uses two scales. A fixed complex neighborhood yields high-order moment information for $X - X'$; after truncation, this produces a Gaussian product approximation on a disk of radius of order $\sqrt{m}$. Zero-freeness on this disk allows one to take a logarithm and recover the factor. The exact theorem, by contrast, follows from equality in Jensen's inequality after spherical averaging.

Exact rigidity

The positive-semidefinite quadratic-form theorem below motivates the reflection condition.

Theorem 1.1 (Exact PSD rigidity) *Let $X = (X_1, \dots, X_d)$ have independent coordinates with $\mathbb{E}X_i = 0$ and $\text{Var}(X_i) = \sigma_i^2$. Let $A = A^\top \succeq 0$ have rank $r \geq 1$, and suppose*

$$X^\top A X \sim \chi_r^2.$$

For every active coordinate i with $A_{ii} > 0$, assume that

$$M_i(s)M_i(-s) \geq e^{\sigma_i^2 s^2}$$

for all s in some neighborhood of the origin. Then every active X_i is Gaussian, $X_i \sim \mathcal{N}(0, \sigma_i^2)$.

No separate MGF-existence hypothesis is required in Theorem 1.1: the chi-square law of the quadratic form itself implies a square-exponential moment for every active coordinate, and therefore finiteness of its MGF on the whole real line; see section 2.

Characterizations based on quadratic forms have a long history. Converse results for sample variance and related forms use additional structural assumptions; see, for example, Christoph–Prohorov–Ulyanov, Kruglov, and Golikova–Kruglov [5, 6, 8]. The condition here is different: infinite divisibility implies the reflected-MGF dominance through the Lévy–Khintchine formula, but the converse fails and no symmetry is required. For an infinitely divisible centered law with finite variance, the representation gives, for every s such that both $M_X(s)$ and $M_X(-s)$ are finite,

$$\log M_X(s) + \log M_X(-s) - \sigma^2 s^2 = 2 \int \left(\cosh(sx) - 1 - \frac{s^2 x^2}{2} \right) \nu(dx) \geq 0.$$

The converse fails. If X takes the values 3 and $-1/3$ with probabilities $1/10$ and $9/10$, then it is centered and has variance one. It is not infinitely divisible: if a bounded non-degenerate law of support diameter d were an n -fold convolution of an i.i.d. root, that root would have support diameter at most d/n , and Popoviciu’s inequality would give $\text{Var}(X)/n \leq d^2/(4n^2)$, which is impossible for large n . Its fourth cumulant is $46/9$, so $R_X(s) = 23s^4/54 + O(s^6)$ and the local dominance holds. This example is asymmetric. In contrast, a Rademacher variable ε satisfies $R_\varepsilon(s) = 2 \log \cosh s - s^2 = -s^4/6 + O(s^6)$, which is negative for all sufficiently small nonzero s .

The Lévy–Khintchine representation used in this comparison is standard; see Sato [9].

Reflection stability

The quantitative result recovers a factor from local reflected-MGF information without assuming the dominance sign.

Theorem 1.2 (Sharp reflection stability) *Fix $\rho > 0$. There exist constants $C_\rho < \infty$ and $\Delta_0(\rho) \in (0, e^{-e})$ such that the following holds. Let X satisfy*

$$\mathbb{E}X = 0, \quad \mathbb{E}X^2 = 1, \quad M_X(s) < \infty \quad (|s| \leq \rho),$$

and suppose

$$|R_X(s)| \leq \Delta, \quad |s| \leq \rho,$$

for some $0 < \Delta \leq \Delta_0(\rho)$. Then

$$d_K(X, \mathcal{N}(0, 1)) \leq C_\rho \sqrt{\frac{\log \log(1/\Delta)}{\log(1/\Delta)}}.$$

Classical Cramér stability begins with a convolution that is already close to a Gaussian convolution in a global probability metric. Sapogov's work is an early cornerstone of this theory [10]; for the i.i.d. equal-summand case substantially stronger estimates are available, for example in Bobkov, Chistyakov, and Götze [11]. In the present problem the input is different: $M_X(s)M_X(-s)$ is controlled only on a fixed real interval, and on the Fourier axis the reflected factorization becomes $|\varphi_X|^2$, which contains no phase information. The main task is therefore to convert local transform control into a global statement about one factor.

One could first try to convert the local reflected-MGF control into a global approximation for $X - X'$ and then invoke a general stability theorem for Cramér decomposition. Such an argument discards the structure responsible for the sharp modulus. Our proof instead keeps the reflected factorization throughout: high-order moment information constructs a zero-free disk for a truncated factor itself, after which the logarithm of that factor is controlled directly. The logarithmic coefficient step is a finite-disk estimate in the spirit of quantitative Marcinkiewicz theory; related connections between zero geometry and Gaussian approximation appear in Eremenko–Fryntov, Dinh–Ghosh–Tran–Tran, and Michelen–Sahasrabudhe [12–14]. The distinctive step here is that the growing zero-free disk is derived from the local reflected-MGF input rather than imposed as a hypothesis.

The order in Theorem 1.2 cannot be improved uniformly. In fact, sharpness already occurs in a symmetric class for which no Fourier phase has to be reconstructed.

Theorem 1.3 (Symmetric Laguerre lower bound) *There are constants $\rho, c, c_-, c_+ > 0$ and centered variance-one random variables W_m with symmetric densities f_m such that, for all sufficiently large m ,*

$$c_- \phi(x) \leq f_m(x) \leq c_+ \phi(x), \quad R_{W_m}(s) \geq 0 \quad (|s| \leq \rho),$$

and, with

$$\Delta_m := \sup_{|s| \leq \rho} R_{W_m}(s) \rightarrow 0,$$

$$\log \frac{1}{\Delta_m} = m \log m + O(m), \quad d_K(W_m, \mathcal{N}(0, 1)) \geq cm^{-1/2}.$$

Consequently, for some $c' > 0$,

$$d_K(W_m, \mathcal{N}(0, 1)) \geq c' \sqrt{\frac{\log \log(1/\Delta_m)}{\log(1/\Delta_m)}}$$

for all sufficiently large m .

Thus the logarithmic instability is already present when $M_X(-s) = M_X(s)$ and the factor on the real axis can be recovered by a positive square root. Reflection-induced phase loss is an additional difficulty for nonsymmetric factor recovery, but it is not the sole source of the optimal modulus.

2 Exact rigidity for chi-square quadratic forms

We prove Theorem 1.1 and then apply it to the ordinary sample variance. Write $A = B^\top B$, where $B \in \mathbb{R}^{r \times d}$ has full row rank, and let b_i be the i th column of B . Thus $A_{ii} = \|b_i\|^2$.

We first note that the chi-square hypothesis automatically gives the MGF regularity needed below. Fix an active coordinate i , so $A_{ii} > 0$, and let

$$Q := X^\top A X \sim \chi_r^2.$$

Independence and centering imply

$$\mathbb{E}(Q \mid X_i) = A_{ii} X_i^2 + \sum_{j \neq i} A_{jj} \sigma_j^2.$$

For every $0 < c < 1/2$, conditional Jensen therefore yields

$$\exp \left(c A_{ii} X_i^2 + c \sum_{j \neq i} A_{jj} \sigma_j^2 \right) \leq \mathbb{E}(e^{cQ} \mid X_i).$$

Taking expectations and using the chi-square MGF gives

$$\mathbb{E} e^{c A_{ii} X_i^2} < \infty. \tag{3}$$

Consequently $M_i(s) < \infty$ for every $s \in \mathbb{R}$, by Young's inequality

$$sx \leq c A_{ii} x^2 + \frac{s^2}{4c A_{ii}}.$$

If V is uniform on S^{r-1} , write

$$\Psi_r(z) := \mathbb{E}_V e^{z V_1}.$$

The transform is well defined for all $z \in \mathbb{C}$ because V_1 is bounded. Spherical averaging gives

$$\mathbb{E}_V \prod_{i=1}^d M_i(t \langle b_i, V \rangle) = \mathbb{E}_{X,V} e^{t \langle BX, V \rangle} = \mathbb{E} \Psi_r(t \sqrt{X^\top A X}). \tag{4}$$

Since $X^\top AX \sim \chi_r^2$, the final quantity in (4) equals

$$\mathbb{E}_{G,V} e^{t\langle G,V \rangle} = e^{t^2/2}, \quad G \sim \mathcal{N}(0, I_r). \quad (5)$$

Only active coordinates enter the product, because $A_{ii} = 0$ implies $b_i = 0$.

For each active coordinate, let the local dominance hold on $|s| < \rho_i$. Since there are finitely many active coordinates, reducing to a common radius $\rho_* := \min_i \rho_i > 0$ makes all subsequent expressions valid for sufficiently small $|t|$.

Define

$$C_i(s) := \log M_i(s) - \frac{\sigma_i^2 s^2}{2}$$

and

$$H_t(v) := \sum_{i=1}^d C_i(t\langle b_i, v \rangle) + \frac{t^2}{2} \left(\sum_{i=1}^d \sigma_i^2 \langle b_i, v \rangle^2 - 1 \right). \quad (6)$$

The quadratic terms cancel inside the exponential, so (5) implies

$$\mathbb{E}_V e^{H_t(V)} = e^{-t^2/2} \mathbb{E}_V \prod_i M_i(t\langle b_i, V \rangle) = 1.$$

Jensen's inequality therefore gives $\mathbb{E}_V H_t(V) \leq 0$.

The mean of the quadratic correction in (6) is zero. Indeed,

$$\mathbb{E}_V \sum_i \sigma_i^2 \langle b_i, V \rangle^2 = \frac{1}{r} \sum_i \sigma_i^2 \|b_i\|^2 = \frac{1}{r} \mathbb{E}(X^\top AX) = 1,$$

where the last equality uses the mean of χ_r^2 . Since V and $-V$ have the same law, coordinatewise dominance now gives

$$\begin{aligned} 2\mathbb{E}_V H_t(V) &= \sum_{i=1}^d \mathbb{E}_V [\log M_i(t\langle b_i, V \rangle) + \log M_i(-t\langle b_i, V \rangle) - \sigma_i^2 t^2 \langle b_i, V \rangle^2] \\ &= \sum_{i=1}^d \mathbb{E}_V R_i(t\langle b_i, V \rangle) \geq 0, \end{aligned}$$

where $R_i(s) = \log M_i(s) + \log M_i(-s) - \sigma_i^2 s^2$. Hence $\mathbb{E}_V H_t(V) = 0$, and every nonnegative summand in the last display has zero expectation.

For an active coordinate, $A_{ii} = \|b_i\|^2 > 0$. If $r \geq 2$, the random variable $\langle b_i, V \rangle$ has a density that is strictly positive on the interior of $[-\|b_i\|, \|b_i\|]$. Continuity of R_i therefore implies $R_i(s) = 0$ on a neighborhood of the origin. If $r = 1$, then $V \in \{-1, 1\}$ and varying t directly gives the same conclusion, because R_i is even. Thus, for every active coordinate,

$$M_i(s)M_i(-s) = e^{\sigma_i^2 s^2}$$

near the origin. If $\sigma_i^2 = 0$, then $X_i = 0$ almost surely. If $\sigma_i^2 > 0$, an independent copy X'_i gives that $X_i - X'_i$ has the MGF of $\mathcal{N}(0, 2\sigma_i^2)$ near zero and hence has that

Gaussian law. The Lévy–Cramér decomposition theorem applied to the independent sum $X_i + (-X'_i)$ then implies that X_i is Gaussian.

This proves Theorem 1.1. For Corollary 5.1, take

$$A = I_n - \frac{1}{n} \mathbf{1}\mathbf{1}^\top.$$

It is a rank- $n - 1$ orthogonal projection, every diagonal entry is $(n - 1)/n > 0$, and its quadratic form is exactly $\sum_i (X_i - \bar{X})^2$. The corollary follows immediately.

3 Reflection stability

Throughout this section let X' be an independent copy of X , set

$$Y = X - X', \quad G \sim \mathcal{N}(0, 2),$$

and define

$$F(z) := M_Y(z) - e^{z^2}, \quad L := \log(1/\Delta).$$

For probability measures μ and ν , write

$$d_{\text{TV}}(\mu, \nu) := \sup_A |\mu(A) - \nu(A)|.$$

All constants in this section may depend on the fixed radius ρ , but not on Δ .

3.1 Automatic exponential integrability

Lemma 3.1 (Local reflected control supplies an exponential envelope) *Under the hypotheses of Theorem 1.2, if $\Delta \leq 1$, then*

$$M_X(s) \leq e^{s^2 + \Delta} \quad (|s| \leq \rho)$$

and

$$\mathbb{E}e^{\rho|X|} \leq M_X(\rho) + M_X(-\rho) \leq 2e^{\rho^2 + 1}. \quad (7)$$

Consequently $\mathbb{E}e^{\rho|Y|}$ is bounded by a constant depending only on ρ .

Proof Since $\mathbb{E}X = 0$, Jensen’s inequality gives $M_X(s) \geq e^{s\mathbb{E}X} = 1$ for every s in the domain of the MGF. Hence $\log M_X(s) \geq 0$. The defect bound gives

$$\log M_X(s) + \log M_X(-s) \leq s^2 + \Delta,$$

so each of the two nonnegative logarithms is at most $s^2 + \Delta$. This proves the first bound. Moreover $e^{\rho|x|} \leq e^{\rho x} + e^{-\rho x}$, which gives (7). Finally,

$$\mathbb{E}e^{\rho|Y|} \leq \mathbb{E}e^{\rho(|X| + |X'|)} = (\mathbb{E}e^{\rho|X|})^2.$$

□

3.2 Half-disk propagation

Lemma 3.2 (Two-constants estimate on a half-disk) *Let $D_R^+ = \{z \in \mathbb{C} : |z| < R, \Im z > 0\}$. Suppose that H is holomorphic on a neighborhood of $\overline{D_R^+}$ and satisfies*

$$|H(x)| \leq a \quad (|x| \leq R), \quad |H(z)| \leq b \quad (|z| = R, \Im z \geq 0),$$

where $0 < a \leq b$. Then for every fixed $0 < \eta < 1$ there is $\omega_\eta > 0$, depending only on η , such that

$$\log |H(z)| \leq \omega_\eta \log a + (1 - \omega_\eta) \log b, \quad |z| \leq \eta R, \quad \Im z \geq 0.$$

The same estimate holds in the lower half-disk.

Proof For $\delta > 0$, the function $u_\delta(z) := \log(|H(z)| + \delta)$ is subharmonic. Its boundary values are at most $\log(a + \delta)$ on the diameter and at most $\log(b + \delta)$ on the semicircle. The two-constants theorem for subharmonic functions therefore bounds u_δ by the corresponding harmonic-measure combination. Under the scaling $z = Rw$, the half-disk D_R^+ becomes the unit half-disk, and the inner region $|z| \leq \eta R$ becomes $|w| \leq \eta$. The harmonic measure of the diameter is positive on this compact inner region, so its minimum is a constant $\omega_\eta > 0$ depending only on η , not on R , m , or Δ . Letting $\delta \downarrow 0$ gives the assertion. $\square$

3.3 High-order moments of the reflected variable

Lemma 3.3 (Local defect to high-order moments) *For every $A > 0$ there is $c_0 = c_0(A, \rho) > 0$ such that, with*

$$m = \left\lfloor c_0 \frac{L}{\log L} \right\rfloor,$$

all sufficiently small Δ satisfy

$$\left| \mathbb{E}Y^k - \mathbb{E}G^k \right| \leq e^{-Am \log m}, \quad 0 \leq k \leq 2m.$$

In particular,

$$\mathbb{E}|Y|^{2m} \leq (Cm)^m.$$

Proof Set $\rho_0 = \rho/2$ and put $D_{\rho_0}^\pm := \{z : |z| < \rho_0, \pm \Im z > 0\}$. By Theorem 3.1, M_Y is holomorphic in the strip $|\Re z| < \rho$, and on the two semicircular arcs $|z| = \rho_0$,

$$|M_Y(z)| \leq \mathbb{E}e^{\rho_0|Y|} \leq C_\rho.$$

Thus $|F(z)| \leq B_0(\rho)$ there. On the real diameter, the defect hypothesis gives

$$|F(s)| = e^{s^2} |e^{Rx(s)} - 1| \leq C_0(\rho)\Delta, \quad |s| \leq \rho_0.$$

For sufficiently small Δ , the diameter bound is no larger than the arc bound. Apply Theorem 3.2 with a fixed $\eta < 1$ to $D_{\rho_0}^+$ and $D_{\rho_0}^-$. For $r_0 := \eta\rho_0$, write $\omega_0 := \omega_\eta$. Thus there are constants $r_0 > 0$, $\omega_0 > 0$, and $C_1 < \infty$, depending only on ρ , such that

$$\sup_{|z| \leq r_0} |F(z)| \leq C_1 \Delta^{\omega_0}.$$

The Cauchy estimate consequently gives

$$|F^{(k)}(0)| \leq C_1 k! r_0^{-k} \Delta^{\omega_0}.$$

For $k \leq 2m$, use $\log(k!) \leq k \log k$ and $k \log k \leq 2m \log(2m)$ to obtain

$$\begin{aligned} \log(C_1 k! r_0^{-k} \Delta^{\omega_0}) &\leq -\omega_0 L + 2m \log(2m) + 2m |\log r_0| + O_\rho(1) \\ &\leq -\omega_0 L + O_\rho(c_0 L), \end{aligned}$$

when $m = \lfloor c_0 L / \log L \rfloor$ and L is large. Since $m \log m \leq c_0 L + o(L)$, choosing c_0 sufficiently small, depending on (A, ρ) , gives $|F^{(k)}(0)| \leq e^{-Am \log m}$. Since $F^{(k)}(0) = \mathbb{E}Y^k - \mathbb{E}G^k$, the moment conclusion follows. Finally,

$$\mathbb{E}G^{2m} = \frac{(2m)!}{m!} \leq (Cm)^m,$$

and the negligible moment mismatch gives the stated $2m$ -th moment bound. $\square$

3.4 Tail bound and truncation

Lemma 3.4 (Square-root tail lifting) *There are constants $C, c > 0$ with the following property. For every sufficiently large fixed B , if $N = B\sqrt{m}$, then*

$$\mathbb{P}(|X| > N) \leq Ce^{-\beta_B m},$$

where $\beta_B \rightarrow \infty$ as $B \rightarrow \infty$. If X_N denotes the conditional law $X \mid \{|X| \leq N\}$, then

$$d_{\text{TV}}(\mathcal{L}(X), \mathcal{L}(X_N)) \leq Ce^{-\beta_B m}.$$

Moreover

$$|\mathbb{E}X_N| + |\text{Var}(X_N) - 1| \leq C(1 + N^2)e^{-cN}.$$

Proof For every $p \geq 1$, conditional Jensen and $\mathbb{E}(Y \mid X) = X$ give

$$|X|^p = |\mathbb{E}(Y \mid X)|^p \leq \mathbb{E}(|Y|^p \mid X),$$

and hence $\mathbb{E}|X|^p \leq \mathbb{E}|Y|^p$. Taking $p = 2m$ and using Theorem 3.3 gives $\|X\|_{2m} \leq C\sqrt{m}$. Therefore, for $N = B\sqrt{m}$,

$$\mathbb{P}(|X| > N) \leq \left(\frac{\|X\|_{2m}}{N} \right)^{2m} \leq C \left(\frac{C_0}{B^2} \right)^m.$$

Thus one may take $\beta_B = \frac{1}{2} \log(B^2/C_0)$ for sufficiently large B . If $p_N = \mathbb{P}(|X| > N)$, conditioning gives

$$d_{\text{TV}}(\mathcal{L}(X), \mathcal{L}(X_N)) \leq p_N.$$

By (7), the exponential envelope implied by the defect bound also gives, for some $c = c(\rho) > 0$,

$$\mathbb{E}(|X| \mathbf{1}_{\{|X| > N\}}) \leq Ce^{-cN}, \quad \mathbb{E}(X^2 \mathbf{1}_{\{|X| > N\}}) \leq C(1 + N^2)e^{-cN}.$$

Since $\mathbb{E}X = 0$ and $\mathbb{E}X^2 = 1$,

$$|\mathbb{E}X_N| \leq \frac{Ce^{-cN}}{1 - p_N}, \quad |\mathbb{E}X_N^2 - 1| \leq \frac{p_N + C(1 + N^2)e^{-cN}}{1 - p_N}.$$

It follows that $|\mathbb{E}X_N| + |\text{Var}(X_N) - 1| \leq C(1 + N^2)e^{-cN}$ after increasing C . $\square$

3.5 Product approximation on a growing interval

Lemma 3.5 (Real-axis product approximation) *For B fixed sufficiently large, there is $a_0 = a_0(B, \rho) > 0$ such that, with*

$$R = a_0 \sqrt{m}$$

and φ_N the characteristic function of X_N ,

$$\sup_{|t| \leq 2R} \left| \varphi_N(t) \varphi_N(-t) - e^{-t^2} \right| \leq e^{-\Gamma m}$$

for the finite target exponent $\Gamma > 4$ selected in Appendix A.

Proof Use the $2m - 1$ Taylor polynomial of the characteristic function of $Y = X - X'$ and the real-variable remainder

$$\left| e^{iu} - \sum_{k=0}^{2m-1} \frac{(iu)^k}{k!} \right| \leq \frac{|u|^{2m}}{(2m)!}.$$

Taylor's inequality and the moment comparison give

$$\begin{aligned} |\varphi_Y(t) - e^{-t^2}| &\leq \frac{|t|^{2m} \mathbb{E}|Y|^{2m}}{(2m)!} + \frac{|t|^{2m} \mathbb{E}|G|^{2m}}{(2m)!} \\ &\quad + \sum_{k=0}^{2m-1} \frac{|t|^k}{k!} |\mathbb{E}(iY)^k - \mathbb{E}(iG)^k|, \end{aligned}$$

where $G \sim \mathcal{N}(0, 2)$. On $|t| \leq 2R$, the two Taylor-remainder terms are bounded by $(Ca_0^2)^m$ after Stirling, while the moment-mismatch term is super-exponentially smaller by Theorem 3.3. Choose a_0 small. Since $\varphi_Y(t) = \varphi_X(t) \varphi_{X'}(-t)$ and the truncation error is bounded by Theorem 3.4, transfer to X_N gives

$$\varphi_N(t) \varphi_N(-t) = \varphi_{X_N - X'_N}(t) = e^{-t^2} + O(e^{-\Gamma m}).$$

The constants in the $O(\cdot)$ term are absorbed by the parameter choices in Appendix A. $\square$

3.6 A zero-free disk

Lemma 3.6 (Growing zero-free disk) *The constants B and a_0 can be chosen so that, for $R = a_0 \sqrt{m}$,*

$$\varphi_N(z) \neq 0, \quad |z| \leq R.$$

More precisely, for some $\gamma_ > a_0^2$,*

$$\sup_{|z| \leq R} \left| \varphi_N(z) \varphi_N(-z) - e^{-z^2} \right| \leq e^{-\gamma_* m}.$$

Proof Since $|X_N| \leq N$, φ_N is entire and

$$|\varphi_N(z)| \leq e^{N|\Im z|}.$$

Hence, for

$$\mathcal{F}_N(z) := \varphi_N(z) \varphi_N(-z) - e^{-z^2},$$

the explicit growth bound

$$\log |\mathcal{F}_N(z)| \leq \log 2 + 4NR + 4R^2 \leq \log 2 + 4(Ba_0 + a_0^2)m, \quad |z| \leq 2R,$$

holds. On the real diameter, Theorem 3.5 gives $\log |\mathcal{F}_N| \leq -\Gamma m$. Apply Theorem 3.2 with radius $2R$ and $\eta = 1/2$ in the upper and lower half-disks. With $\omega_0 := \omega_{1/2}$, this gives

$$\log |\mathcal{F}_N(z)| \leq -\gamma_{\text{raw}} m + O(1), \quad \gamma_{\text{raw}} := \omega_0 \Gamma - 4(1 - \omega_0)(Ba_0 + a_0^2).$$

Appendix A chooses B and then a_0 so that $\gamma_{\text{raw}} > a_0^2$. Choose once and for all

$$a_0^2 < \gamma_* < \gamma_{\text{raw}}.$$

For all sufficiently large m , the $O(1)$ term is then absorbed and

$$|\mathcal{F}_N(z)| \leq e^{-\gamma_* m}, \quad |z| \leq R.$$

Since

$$\inf_{|z| \leq R} |e^{-z^2}| \geq e^{-R^2} = e^{-a_0^2 m},$$

we have

$$\frac{|\mathcal{F}_N(z)|}{|e^{-z^2}|} \leq e^{-(\gamma_* - a_0^2)m} < \frac{1}{2}$$

for all sufficiently large m . Hence

$$|\varphi_N(z)\varphi_N(-z)| \geq \frac{1}{2}|e^{-z^2}| > 0,$$

and therefore $\varphi_N(z) \neq 0$ on the disk. $\square$

3.7 From zero-freeness to Gaussian approximation

Lemma 3.7 (From zero-freeness to Gaussian approximation) *Under the conclusions of Theorem 3.6,*

$$d_K(X_N, \mathcal{N}(\mu_N, \sigma_N^2)) \leq \frac{C}{R},$$

where $\mu_N = \mathbb{E}X_N$ and $\sigma_N^2 = \text{Var}(X_N)$. Consequently

$$d_K(X, \mathcal{N}(0, 1)) \leq \frac{C}{\sqrt{m}}.$$

Proof The disk $D_R := \{z : |z| < R\}$ is simply connected, and Theorem 3.6 gives $\varphi_N \neq 0$ there. Hence there is a unique analytic logarithm on D_R satisfying $\psi_N(0) = 0$; write

$$\psi_N(z) = \log \varphi_N(z), \quad \psi_N(0) = 0.$$

The bounded support gives $\Re \psi_N(z) = \log |\varphi_N(z)| \leq NR$. Set

$$H(z) := NR - \psi_N(z), \quad \Re H(z) \geq 0.$$

After scaling $z = Rw$, the Carathéodory coefficient theorem for functions with nonnegative real part (see, for example, Duren [15]) applied to $H(Rw)/(NR)$ gives, for $\psi_N(z) = \sum_{k \geq 1} a_k z^k$,

$$|a_k| \leq 2NR^{1-k}.$$

Since $a_1 = i\mu_N$ and $a_2 = -\sigma_N^2/2$, for every fixed $0 < \theta < 1$,

$$\left| \psi_N(t) - i\mu_N t + \frac{\sigma_N^2 t^2}{2} \right| \leq \sum_{k \geq 3} 2NR^{1-k} |t|^k \leq \frac{2N|t|^3}{R^2(1 - |t|/R)} \leq C \frac{|t|^3}{R}, \quad |t| \leq \theta R,$$

because $N/R = B/a_0$ is fixed. Define the cubic remainder

$$E_N(t) := \psi_N(t) - i\mu_N t + \frac{\sigma_N^2 t^2}{2}.$$

By Theorem 3.4, for all sufficiently large m ,

$$|\mu_N| = o(R^{-1}), \quad \frac{1}{2} \leq \sigma_N^2 \leq \frac{3}{2}, \quad |E_N(t)| \leq C \frac{|t|^3}{R}.$$

Choose θ so small that $C\theta \leq 1/8$. Then, on $|t| \leq \theta R$,

$$\Re \psi_N(t) = -\frac{\sigma_N^2 t^2}{2} + \Re E_N(t) \leq -\frac{t^2}{4} + \frac{t^2}{8} = -\frac{t^2}{8}.$$

Writing

$$\varphi_N(t) = e^{i\mu_N t - \sigma_N^2 t^2/2} e^{E_N(t)},$$

and using $|e^z - 1| \leq |z|e^{|z|}$ gives Gaussian damping in the difference:

$$\left| \varphi_N(t) - e^{i\mu_N t - \sigma_N^2 t^2/2} \right| \leq C \frac{|t|^3}{R} e^{-ct^2}.$$

Let g_N be the density of $\mathcal{N}(\mu_N, \sigma_N^2)$. The Esseen smoothing inequality in the present normalization (Esseen [16]) is

$$\begin{aligned} d_K(X_N, \mathcal{N}(\mu_N, \sigma_N^2)) &\leq \frac{1}{\pi} \int_{-T}^T \left| \frac{\varphi_N(t) - e^{i\mu_N t - \sigma_N^2 t^2/2}}{t} \right| dt + \frac{24\|g_N\|_\infty}{\pi T} \\ &= \frac{1}{\pi} \int_{-T}^T \left| \frac{\varphi_N(t) - e^{i\mu_N t - \sigma_N^2 t^2/2}}{t} \right| dt + O(T^{-1}), \end{aligned}$$

where $\|g_N\|_\infty = (\sqrt{2\pi}\sigma_N)^{-1}$ and the last form uses $\sigma_N^2 \geq 1/2$. With $T = \theta R$, the integral is bounded by

$$\frac{C}{R} \int_{-T}^T t^2 e^{-ct^2} dt \leq \frac{C}{R} \int_{\mathbb{R}} t^2 e^{-ct^2} dt = O(R^{-1}),$$

and the cutoff term is also $O(R^{-1})$.

It remains to compare the nearby Gaussian with the standard one. For σ in a fixed compact subset of $(0, \infty)$,

$$d_K(\mathcal{N}(\mu, \sigma^2), \mathcal{N}(0, 1)) \leq C(|\mu| + |\sigma - 1|). \quad (8)$$

Indeed, the translation term is bounded by $C|\mu|$ using the bounded Gaussian density, while the mean-value theorem in σ gives

$$\sup_x |\Phi(x/\sigma) - \Phi(x)| \leq C|\sigma - 1|,$$

because $\sup_y |y|\phi(y) < \infty$. Since $\sigma_N^2 \in [1/2, 3/2]$, also $|\sigma_N - 1| \leq C|\sigma_N^2 - 1|$. Thus Theorem 3.4 and (8) show that the Gaussian parameter shift is $o(R^{-1})$. Finally,

$$d_K(X, X_N) \leq d_{TV}(\mathcal{L}(X), \mathcal{L}(X_N)) = o(R^{-1}),$$

so the triangle inequality yields the consequence for X . $\square$

Proof of Theorem 1.2 First apply Theorem 3.1; this is exactly where the exponential envelope formerly imposed as a separate hypothesis is obtained from the reflected defect. Then apply Theorems 3.3 to 3.7. Since

$$m \asymp_\rho \frac{L}{\log L}, \quad L = \log(1/\Delta),$$

we have

$$\frac{1}{\sqrt{m}} \asymp_\rho \sqrt{\frac{\log L}{L}},$$

which is the claimed rate. $\square$

4 Optimality and obstruction results

4.1 A symmetric Laguerre lower-bound family

The sharp modulus does not require an asymmetric perturbation. We construct a symmetric family whose densities are uniformly comparable with the Gaussian density and for which the reflected defect is nonnegative. The only analytic input is the following uniform Laguerre envelope; its normalization and four-region proof are recorded in Appendix B.

Lemma 4.1 (Uniform even Laguerre envelope)

$$C_E := \sup_{m \geq 2, x \in \mathbb{R}} \sqrt{m} e^{-x^2} \left| L_m^{(-1/2)}(3x^2/2) \right| < \infty. \quad (9)$$

Proof See Appendix B. □

Fix $0 < \epsilon < 1/4$, let ϕ be the standard Gaussian density, and define, for $m \geq 2$,

$$h_m(x) := \frac{(-1)^m \sqrt{m}}{C_E} e^{-x^2} L_m^{(-1/2)}(3x^2/2), \quad f_m(x) := \phi(x)(1 + \epsilon h_m(x)).$$

By (9), $|h_m| \leq 1$, so

$$(1 - \epsilon)\phi(x) \leq f_m(x) \leq (1 + \epsilon)\phi(x). \quad (10)$$

In particular f_m is nonnegative and has exponential moments of every fixed order uniformly in m .

The perturbation is even. With $r = 3x^2/2$, Laguerre orthogonality for parameter $-1/2$ gives, for $0 \leq k < m$,

$$\int_{\mathbb{R}} x^{2k} h_m(x) \phi(x) dx = C_{m,k} \int_0^\infty r^{k-1/2} e^{-r} L_m^{(-1/2)}(r) dr = 0,$$

where $C_{m,k} \neq 0$ is an irrelevant deterministic factor. The cases $k = 0$ and $k = 1$ show that f_m has total mass one and variance one, while evenness gives mean zero. Let W_m have density f_m .

To compute the MGF exactly, set

$$J_m(s) := \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} e^{sx - 3x^2/2} L_m^{(-1/2)}(3x^2/2) dx.$$

The Laguerre generating function gives, for $|t| < 1$,

$$\begin{aligned} \sum_{m \geq 0} J_m(s) t^m &= \frac{(1-t)^{-1/2}}{\sqrt{2\pi}} \int_{\mathbb{R}} \exp\left(sx - \frac{3x^2}{2(1-t)}\right) dx \\ &= \frac{1}{\sqrt{3}} \exp\left(\frac{s^2(1-t)}{6}\right) \end{aligned}$$

$$= \frac{1}{\sqrt{3}} e^{s^2/6} \sum_{m \geq 0} \frac{(-s^2/6)^m}{m!} t^m.$$

Thus

$$J_m(s) = \frac{1}{\sqrt{3}} e^{s^2/6} \frac{(-s^2/6)^m}{m!}.$$

After multiplying by the factor $(-1)^m \sqrt{m}/C_E$ in h_m and dividing by the Gaussian MGF $e^{s^2/2}$, we obtain

$$M_{W_m}(s) = e^{s^2/2} (1 + v_m(s)), \quad v_m(s) := \frac{\epsilon \sqrt{m}}{\sqrt{3} C_E} e^{-s^2/3} \frac{(s^2/6)^m}{m!}. \quad (11)$$

In particular $v_m(s) \geq 0$ on the real axis. Since W_m is symmetric,

$$R_{W_m}(s) = 2 \log(1 + v_m(s)) \geq 0. \quad (12)$$

Fix $\rho = 1$. For $m \geq 1$, $s \mapsto e^{-s^2/3} s^{2m}$ is increasing on $[0, 1]$, so

$$\Delta_m := \sup_{|s| \leq 1} R_{W_m}(s) = 2 \log(1 + v_m(1)).$$

Since $v_m(1) \rightarrow 0$, we have $\Delta_m \asymp v_m(1)$, and Stirling's formula gives

$$\log \frac{1}{\Delta_m} = m \log m + O(m). \quad (13)$$

Hence

$$\sqrt{\frac{\log \log(1/\Delta_m)}{\log(1/\Delta_m)}} \asymp m^{-1/2}.$$

It remains to prove a Kolmogorov discrepancy of this order. The value at the origin is

$$L_m^{(-1/2)}(0) = \frac{(1/2)_m}{m!} = \frac{1}{4^m} \binom{2m}{m} \asymp m^{-1/2}.$$

Moreover,

$$\frac{L_m^{(-1/2)}(r)}{L_m^{(-1/2)}(0)} = \sum_{k=0}^m \frac{(-m)_k}{(1/2)_k k!} r^k. \quad (14)$$

Choose a fixed $c_0 > 0$ so small that $e^{2c_0} - 1 \leq 1/2$. For $0 \leq r \leq c_0/m$, using $|(-m)_k| \leq m^k$ and $(1/2)_k \geq 2^{-k} k!$ gives

$$\left| \frac{L_m^{(-1/2)}(r)}{L_m^{(-1/2)}(0)} - 1 \right| \leq \sum_{k \geq 1} \frac{(2mr)^k}{(k!)^2} \leq e^{2c_0} - 1 \leq \frac{1}{2}.$$

Therefore $L_m^{(-1/2)}(r)$ keeps the sign of its value at the origin and is at least half as large there. Put

$$x_m := \sqrt{\frac{2c_0}{3m}}.$$

For $0 \leq x \leq x_m$, (9) and the preceding lower bound show that $h_m(x)$ has the fixed sign $(-1)^m$ and

$$|h_m(x)| \geq c_1$$

with $c_1 > 0$ independent of m . Since $f_m - \phi = \epsilon h_m \phi$ is even and has total integral zero, its integral over $(-\infty, 0]$ vanishes. Consequently

$$\begin{aligned} d_K(W_m, \mathcal{N}(0, 1)) &\geq |F_{W_m}(x_m) - \Phi(x_m)| \\ &= \epsilon \left| \int_0^{x_m} h_m(x) \phi(x) dx \right| \geq c_2 x_m \geq c_3 m^{-1/2}. \end{aligned}$$

Together with (13), this proves Theorem 1.3. In particular no uniform power-law modulus $C\Delta^\alpha$, $\alpha > 0$, can replace the logarithmic modulus in Theorem 1.2.

The construction is symmetric, so on the real axis $M_{W_m}(-s) = M_{W_m}(s)$ and there is no phase ambiguity in recovering the MGF factor from its reflected product. Thus the matching lower bound isolates a more basic source of instability: recovering a global distribution from extremely accurate transform information on only a fixed real interval.

4.2 Failure of total-variation stability

Throughout this subsection, $d_{TV}(\mu, \nu) := \sup_A |\mu(A) - \nu(A)|$.

Proposition 4.2 (Strong-distance obstruction) *There is no uniform total-variation stability theorem under the assumptions of Theorem 1.2: a standardized Poisson lattice family has reflected defect tending to zero on every fixed interval while its total-variation distance from every absolutely continuous Gaussian law equals one.*

Let $N_\lambda \sim \text{Poisson}(\lambda)$ and $X_\lambda = (N_\lambda - \lambda)/\sqrt{\lambda}$. Its centered cumulant generating function is

$$K_\lambda(s) = \lambda \left(e^{s/\sqrt{\lambda}} - 1 - \frac{s}{\sqrt{\lambda}} \right).$$

Consequently, uniformly on every fixed compact s -interval,

$$\begin{aligned} R_{X_\lambda}(s) &= K_\lambda(s) + K_\lambda(-s) - s^2 \\ &= 2\lambda \left(\cosh(s/\sqrt{\lambda}) - 1 - \frac{s^2}{2\lambda} \right) = \frac{s^4}{12\lambda} + O_\rho(\lambda^{-2}). \end{aligned}$$

For fixed $b > 0$, the inequality $e^u - 1 - u \leq u^2 e^{|u|}/2$ gives

$$\mathbb{E} e^{b|X_\lambda|} \leq \mathbb{E} e^{bX_\lambda} + \mathbb{E} e^{-bX_\lambda} \leq 2 \exp \left(\frac{b^2}{2} e^{b/\sqrt{\lambda}} \right),$$

so the family even has a fixed exponential-integrability bound for each fixed b . Nevertheless X_λ is lattice-valued, whereas $\mathcal{N}(0, 1)$ is absolutely continuous, so

$$d_{\text{TV}}(\mathcal{L}(X_\lambda), \mathcal{N}(0, 1)) = 1.$$

This is a qualitative boundary phenomenon and is independent of the Kolmogorov lower bound above.

5 Sample variance

Let $X_1, \dots, X_n$ be i.i.d. copies of a centered variance-one random variable X , let $\bar{X} = n^{-1} \sum_i X_i$, and set

$$Q_n = \sum_{i=1}^n (X_i - \bar{X})^2.$$

For $r \geq 1$, let V be uniform on S^{r-1} . Recall the spherical transform

$$\Psi_r(z) := \mathbb{E}_V e^{zV_1}.$$

Define

$$\mathfrak{D}_{r,\tau}(Q) := \sup_{|t| \leq \tau} \left| e^{-t^2/2} \mathbb{E} \Psi_r(t\sqrt{Q}) - 1 \right|.$$

This is the spherical Laplace observable associated with a squared radial projection; it is not itself a probability metric on the law of Q . Below we first derive the one-dimensional reflected defect from this observable and then show that, under the same exponential envelope used in the quantitative sample-variance theorem, ordinary Kolmogorov distance from the chi-square law controls the observable.

The exact sample-variance consequence of the quadratic-form theorem is the following.

Corollary 5.1 (Exact sample-variance characterization under dominance) *Let $n \geq 2$ and let $X_1, \dots, X_n$ be i.i.d., centered, and variance one. If*

$$M_X(s)M_X(-s) \geq e^{s^2}$$

for all s in some neighborhood of the origin, and

$$\sum_{i=1}^n (X_i - \bar{X})^2 \sim \chi_{n-1}^2,$$

then $X_i \sim \mathcal{N}(0, 1)$.

As in Theorem 1.1, no separate MGF-existence assumption is needed: the chi-square law of the quadratic form implies it automatically for every coordinate.

The quantitative statement based on the spherical observable is the next theorem.

Theorem 5.2 (Sample-variance stability under spherical Laplace discrepancy) *Fix $n \geq 3$, $\tau, K > 0$, and $0 < \rho' < \tau\sqrt{(n-1)/n}$. There are constants $C < \infty$ and $\varepsilon_0 > 0$, depending only on (n, τ, K, ρ') , such that the following holds. Let $X_1, \dots, X_n$ be i.i.d. copies of a centered variance-one X with $\mathbb{E}e^{2\tau|X|} \leq K$, and assume $R_X(s) \geq 0$ for $|s| \leq \tau$. If*

$$Q_n = \sum_{i=1}^n (X_i - \bar{X})^2, \quad \mathfrak{D}_{n-1, \tau}(Q_n) \leq \varepsilon \leq \varepsilon_0,$$

then

$$d_K(X, \mathcal{N}(0, 1)) \leq C \sqrt{\frac{\log \log(1/\varepsilon)}{\log(1/\varepsilon)}}.$$

The discrepancy-to-defect transfer is as follows. Let $r = n - 1$, let $P = I_n - n^{-1}\mathbf{1}\mathbf{1}^\top$, and choose $B \in \mathbb{R}^{r \times n}$ with $B^\top B = P$ and $BB^\top = I_r$. If V is uniform on S^{r-1} , then every column b_i of B has norm

$$q := \|b_i\| = \sqrt{\frac{n-1}{n}},$$

and $\langle b_i, V \rangle$ has the same distribution as qV_1 . Put $a = \tau q$ and define

$$D_X(s) := \frac{1}{2}R_X(s) = \frac{1}{2} \log \frac{M_X(s)M_X(-s)}{e^{s^2}}.$$

Under dominance, $D_X \geq 0$ on $[-\tau, \tau]$.

The exact spherical transform gives, for $|t| \leq \tau$,

$$e^{-t^2/2} \mathbb{E} \Psi_r(t\sqrt{Q_n}) = \mathbb{E}_V \exp \left(\sum_{i=1}^n \left[\log M_X(t\langle b_i, V \rangle) - \frac{t^2 \langle b_i, V \rangle^2}{2} \right] \right). \quad (15)$$

Jensen's inequality, exchangeability, and the symmetry of V_1 therefore imply

$$n \mathbb{E} D_X(aV_1) \leq \log(1 + \mathfrak{D}_{n-1, \tau}(Q_n)). \quad (16)$$

Equation (16) connects the spherical observable to one-dimensional reflection control.

For the pointwise transfer, the density of V_1 is

$$f_n(v) = \frac{\Gamma((n-1)/2)}{\sqrt{\pi} \Gamma((n-2)/2)} (1-v^2)^{(n-4)/2}, \quad -1 < v < 1.$$

Fix $0 < \rho' < a$ and choose $0 < h < a - \rho'$. On $[-(\rho' + h), \rho' + h]$ the scaled density of aV_1 is bounded below by

$$m_h := \frac{1}{a} \frac{\Gamma((n-1)/2)}{\sqrt{\pi} \Gamma((n-2)/2)} \left(1 - \left(\frac{\rho' + h}{a} \right)^2 \right)^{\max(n-4, 0)/2}.$$

The exponential envelope implies that D_X is Lipschitz there. One may take

$$L_h := \frac{K}{e\tau} + \rho' + h,$$

because $M_X(s) \geq 1$ and $|x|e^{\tau|x|} \leq (e\tau)^{-1}e^{2\tau|x|}$. If $M_* := \sup_{|s| \leq \rho'} D_X(s)$, choose $s_0 \in [-\rho', \rho']$ with $D_X(s_0) = M_*$. The function D_X is even, nonnegative, and satisfies $D_X(0) = 0$. Since it is L_h -Lipschitz on the enlarged interval,

$$D_X(s) \geq (M_* - L_h|s - s_0|)_+.$$

Moreover, $|s_0| \geq M_*/L_h$, so the analogous tent centered at $-s_0$ has disjoint interior from the other one. Integrating the two tents against the density lower bound gives

$$2m_h I_{L_h, h}(M_*) \leq \frac{1}{n} \log(1 + \mathfrak{D}_{n-1, \tau}(Q_n)),$$

where

$$I_{L, h}(M) := \begin{cases} M^2/L, & M \leq Lh, \\ 2hM - Lh^2, & M > Lh. \end{cases}$$

When $M \leq Lh$, the integral is the area of a triangle of base $2M/L$; when $M > Lh$, it is the trapezoid obtained after the tent reaches the edge of the available interval. Writing $G(y; L, h) = \sqrt{Ly}$ when $y \leq Lh^2$ and $G(y; L, h) = y/(2h) + Lh/2$ otherwise, we obtain

$$\sup_{|s| \leq \rho'} D_X(s) \leq \inf_{0 < h < a - \rho'} G\left(\frac{\log(1 + \mathfrak{D}_{n-1, \tau}(Q_n))}{2nm_h}; L_h, h\right). \quad (17)$$

In particular, for fixed n, τ, K, ρ' , the right-hand side is $O(\sqrt{\mathfrak{D}_{n-1, \tau}(Q_n)})$ as the discrepancy tends to zero. The strict interior restriction is used by this density-lower-bound argument when $n \geq 5$, because the spherical density vanishes at the endpoints; an endpoint result would require a different local estimate.

Applying (17) to the sample-variance hypothesis gives

$$\Delta_{\rho'}(X) \leq C_1 \sqrt{\varepsilon},$$

for all sufficiently small ε , where $C_1 = C_1(n, \tau, K, \rho')$; here we used $R_X = 2D_X$. Set $\Delta = C_1 \sqrt{\varepsilon}$. Then

$$\log(1/\Delta) = \frac{1}{2} \log(1/\varepsilon) - \log C_1 \asymp \log(1/\varepsilon), \quad \log \log(1/\Delta) = \log \log(1/\varepsilon) + O(1).$$

Theorem 1.2, with radius ρ' , therefore yields the displayed rate in Theorem 5.2. Notice that the exponential envelope in Theorem 5.2 is used here to transfer the spherical average to a pointwise defect; it is no longer needed as an independent hypothesis of Theorem 1.2 itself.

We next relate the spherical observable to an ordinary probability metric. Let $H_r \sim \chi_r^2$.

Lemma 5.3 (Kolmogorov control of the spherical observable) *Fix $r \geq 1$ and $\tau > 0$. Let $Q \geq 0$ and suppose*

$$\mathbb{E}e^{2\tau\sqrt{Q}} \leq A.$$

Then, for

$$\delta := d_K(Q, H_r) \leq 1,$$

one has

$$\mathfrak{D}_{r,\tau}(Q) \leq C_{r,\tau,A}\sqrt{\delta}.$$

Proof For $|t| \leq \tau$, define

$$g_t(q) := e^{-t^2/2}\Psi_r(t\sqrt{q}), \quad q \geq 0.$$

Because V_1 is symmetric,

$$\Psi_r(t\sqrt{q}) = \mathbb{E}_V \cosh(t\sqrt{q} V_1),$$

so g_t is nonnegative and nondecreasing in q . Also

$$0 \leq g_t(q) \leq e^{|t|\sqrt{q}} \leq e^{\tau\sqrt{q}}.$$

If $H_r = \|G\|^2$ with $G \sim \mathcal{N}(0, I_r)$, spherical symmetry gives

$$\mathbb{E}g_t(H_r) = e^{-t^2/2}\mathbb{E}e^{tG_1} = 1.$$

Moreover,

$$\sup_{|t| \leq \tau} \mathbb{E}g_t(Q)^2 \leq A, \quad \sup_{|t| \leq \tau} \mathbb{E}g_t(H_r)^2 \leq \mathbb{E}e^{2\tau\sqrt{H_r}} =: A_{r,\tau} < \infty.$$

For $T \geq 1$, the function $g_t \wedge T$ is bounded and nondecreasing, with total variation at most T . Integration by parts against the difference of the distribution functions therefore gives

$$|\mathbb{E}(g_t(Q) \wedge T) - \mathbb{E}(g_t(H_r) \wedge T)| \leq T\delta.$$

The tails satisfy

$$\mathbb{E}(g_t(Q) - T)_+ \leq \frac{\mathbb{E}g_t(Q)^2}{T} \leq \frac{A}{T},$$

and likewise for H_r . Hence, uniformly in $|t| \leq \tau$,

$$|\mathbb{E}g_t(Q) - 1| \leq T\delta + \frac{A + A_{r,\tau}}{T}.$$

Taking $T = \delta^{-1/2}$ proves the claim. $\square$

For the sample variance, the exponential envelope already assumed in Theorem 5.2 provides the hypothesis of Theorem 5.3. Indeed, since $Q_n = X^\top P X \leq \sum_i X_i^2$,

$$\sqrt{Q_n} \leq \left(\sum_i X_i^2 \right)^{1/2} \leq \sum_i |X_i|,$$

and independence gives

$$\mathbb{E}e^{2\tau\sqrt{Q_n}} \leq \prod_{i=1}^n \mathbb{E}e^{2\tau|X_i|} \leq K^n. \quad (18)$$

We therefore obtain the following standard-distance consequence.

Corollary 5.4 (Sample-variance stability in Kolmogorov distance) *Fix $n \geq 3$, $\tau, K > 0$, and $0 < \rho' < \tau\sqrt{(n-1)/n}$. There exist constants $C < \infty$ and $\delta_0 > 0$, depending only on (n, τ, K, ρ') , such that the following holds. Let $X_1, \dots, X_n$ be i.i.d. copies of a centered variance-one X satisfying*

$$\mathbb{E}e^{2\tau|X|} \leq K, \quad R_X(s) \geq 0 \quad (|s| \leq \tau).$$

If

$$\delta := d_K(Q_n, \chi_{n-1}^2) \leq \delta_0,$$

then

$$d_K(X, \mathcal{N}(0, 1)) \leq C \sqrt{\frac{\log \log(1/\delta)}{\log(1/\delta)}}.$$

Proof By (18) and Theorem 5.3,

$$\mathfrak{D}_{n-1, \tau}(Q_n) \leq C_0 \sqrt{\delta}.$$

Apply Theorem 5.2 with $\varepsilon = C_0 \sqrt{\delta}$. Since

$$\log(1/\varepsilon) = \frac{1}{2} \log(1/\delta) + O(1), \quad \log \log(1/\varepsilon) = \log \log(1/\delta) + O(1),$$

the asserted rate follows after changing the constant. $\square$

In the exact case, Theorem 5.1 gives Gaussianity under the same dominance condition. For $n = 2$, the exact corollary is covered by the rank-one case of Theorem 1.1: $A = I_2 - \frac{1}{2}\mathbf{1}\mathbf{1}^\top$ has rank one and both diagonal entries are positive. Thus the discrepancy-to-defect argument is a quantitative extension of the exact spherical-Jensen mechanism.

Several questions remain. It would be useful to determine whether the Kolmogorov-distance modulus in Theorem 5.4 is sharp, and whether the exponential envelope used to pass from a probability metric on Q_n to the unbounded spherical Laplace observable can be weakened. Another direction is to replace local real-axis MGF control by weaker transform or moment information while retaining the optimal reflection modulus. Finally, a corresponding quantitative theory for general positive-semidefinite quadratic forms with non-identically distributed coordinates remains open.

A Parameter selection for the sharp upper bound

This appendix records a non-circular order of choices for the proof of Theorem 1.2. Fix $\rho > 0$ and the interior analytic radius $\rho/2$ used in Theorem 3.3. Let $\omega_0 > 0$ be the lower bound for the harmonic measure of the real diameter in the smaller half-disk used in the propagation step. Choose $A > 8$ and then choose $c_0 = c_0(A, \rho) > 0$ sufficiently small that the Cauchy estimate in Theorem 3.3 dominates all factorial losses; equivalently, after absorbing the fixed ρ -dependent constants, one may require

$$(A + 3)c_0 < \omega_0.$$

With $m = \lfloor c_0 L / \log L \rfloor$, this gives the required super-exponential moment accuracy.

Fix a finite target exponent $\Gamma > 4$. Choose the truncation factor B sufficiently large that the tail-lifting estimate has exponent $\beta_B > 4\Gamma$. After B is fixed, let C_T denote the constant in the real-axis Taylor remainder and choose $a_0 > 0$ so small that

$$C_T a_0^2 < e^{-2\Gamma}, \quad 4(1 - \omega_0)(Ba_0 + a_0^2) < \frac{\omega_0 \Gamma}{2}, \quad a_0^2 < \frac{\omega_0 \Gamma}{4}, \quad -\log(C_T a_0^2) > 4\Gamma.$$

The two Taylor-remainder contributions are then bounded by $e^{-\Gamma m}$ after absorbing fixed multiplicative constants, while the truncation contribution is also bounded by $e^{-\Gamma m}$ because $\beta_B > 4\Gamma$. Thus Theorem 3.5 uses the target exponent Γ , while the complex propagation estimate has raw exponent

$$\gamma_{\text{raw}} := \omega_0 \Gamma - 4(1 - \omega_0)(Ba_0 + a_0^2) \geq \frac{\omega_0 \Gamma}{2} > a_0^2.$$

Choose any fixed γ_* with

$$a_0^2 < \gamma_* < \gamma_{\text{raw}}.$$

The $O(1)$ term in the two-constants estimate is then absorbed for all sufficiently large m , giving the exponent used in Theorem 3.6. Finally choose the fixed Esseen fraction $0 < \theta < 1/4$ so that the cubic remainder in the logarithmic expansion is at most one quarter of the Gaussian quadratic damping on $|t| \leq \theta R$.

The choices are made in the order

$$A \rightarrow c_0 \rightarrow \Gamma \rightarrow B \rightarrow a_0 \rightarrow \gamma_* \rightarrow \theta,$$

and no later choice affects an earlier inequality.

B A uniform Laguerre estimate

We prove the envelope used in Theorem 4.1. For $\alpha = -1/2$, introduce the normalized Laguerre function

$$\mathcal{L}_m^{(-1/2)}(r) = a_m L_m^{(-1/2)}(r) e^{-r/2} r^{-1/4}, \quad a_m = \sqrt{\frac{m!}{\Gamma(m + 1/2)}}, \quad \nu = 4m + 1.$$

Proposition 3.2 of Imekraz, Robert, and Thomann [17] applies for every $\alpha > -1$, hence in particular for $\alpha = -1/2$. It gives constants $C, c > 0$, independent of m and r , such that

$$\begin{aligned} |\mathcal{L}_m^{(-1/2)}(r)| &\leq C(r\nu)^{-1/4} \quad (0 \leq r \leq \nu^{-1}), \\ |\mathcal{L}_m^{(-1/2)}(r)| &\leq C(r\nu)^{-1/4} \quad (\nu^{-1} \leq r \leq \nu/2), \\ |\mathcal{L}_m^{(-1/2)}(r)| &\leq C\nu^{-1/4}(\nu^{1/3} + |\nu - r|)^{-1/4} \quad (\nu/2 \leq r \leq 3\nu/2), \end{aligned}$$

and

$$|\mathcal{L}_m^{(-1/2)}(r)| \leq Ce^{-cr} \quad (r \geq 3\nu/2).$$

Here

$$a_m^{-1} \asymp m^{-1/4}, \quad \sqrt{m} a_m^{-1} \asymp m^{1/4} \asymp \nu^{1/4}.$$

Since

$$L_m^{(-1/2)}(r) = a_m^{-1} \mathcal{L}_m^{(-1/2)}(r) e^{r/2} r^{1/4},$$

we need to bound

$$P_m(r) := \sqrt{m} e^{-2r/3} |L_m^{(-1/2)}(r)| = \sqrt{m} a_m^{-1} r^{1/4} e^{-r/6} |\mathcal{L}_m^{(-1/2)}(r)|.$$

In the first two regions, $0 \leq r \leq \nu/2$, the common estimate gives

$$P_m(r) \leq C\nu^{1/4} r^{1/4} (r\nu)^{-1/4} e^{-r/6} \leq C.$$

In the turning region, $\nu/2 \leq r \leq 3\nu/2$,

$$P_m(r) \leq Cr^{1/4} (\nu^{1/3} + |\nu - r|)^{-1/4} e^{-r/6} \leq C\nu^{1/6} e^{-\nu/12} \leq C.$$

Finally, for $r \geq 3\nu/2$,

$$P_m(r) \leq C\nu^{1/4} r^{1/4} e^{-(c+1/6)r} \leq C.$$

Thus

$$\sup_{m \geq 2, r \geq 0} \sqrt{m} e^{-2r/3} |L_m^{(-1/2)}(r)| < \infty.$$

Substituting $r = 3x^2/2$ gives exactly

$$\sup_{m \geq 2, x \in \mathbb{R}} \sqrt{m} e^{-x^2} |L_m^{(-1/2)}(3x^2/2)| < \infty,$$

which is (9).

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